

Vibe Coding – Chat History

Lab 4: VibeCoding

Only the Vibe Coding interactions for the four Lunar Lander tasks are included.

Task 1 — Fix Horizontal Landing Speed

Vibe Coding prompt

Fix the touchdown/landing validation in game.py. A landing should only be successful if the ship is over a landing pad, absolute horizontal velocity (self.vel.x) is $\leq$ MAX_SPEED_X, absolute vertical velocity (self.vel.y) is $\leq$ MAX_SPEED_Y, and absolute landing angle is $\leq$ MAX_ANGLE. Do not change other functionality or add new features.

Result / implementation

The landing validation was changed to check horizontal speed, vertical speed, and angle.

Relevant code

```
ok_x = abs(self.vel.x) <= MAX_SPEED_X
ok_y = abs(self.vel.y) <= MAX_SPEED_Y
ok_a = abs(angle) <= MAX_ANGLE
if pad and ok_x and ok_y and ok_a:
```

Task 2 — Fuel-Based Ship Colour

Vibe Coding prompt

Implement ship_color(fuel_ratio) in game.py. High fuel should be green, medium fuel yellow, and low fuel red. Use fuel_ratio where 1.0 is full and 0.0 is empty. Integrate it with the existing drawing code. Do not modify unrelated functionality.

Result / implementation

The function clamps fuel_ratio and returns green, yellow, or red according to the remaining fuel.

Relevant code

```
ratio = max(0.0, min(1.0, fuel_ratio))
if ratio > 0.6:
    return (80, 220, 90)
if ratio > 0.2:
    return (220, 200, 50)
return (220, 70, 70)
```

Task 3 — Landing Fireworks

Vibe Coding prompt

Implement on_landing(score) in game.py. A successful landing should display a visible fireworks/particle effect for a short period using pygame. It should trigger only after successful landing and not interfere with normal gameplay. Do not modify unrelated functionality.

Result / implementation

A pygame particle/fireworks system was added. Successful landings spawn particles at the landing position, which are updated and drawn during the game loop.

Relevant code

```
def _spawn_fireworks(pos):
    colors = [(255, 80, 80), (90, 200, 255), (255, 220, 90), (150, 255, 120), (255, 150, 240)]
```

Task 4 — Bonus Life at 1500 Points

Vibe Coding prompt

Implement `bonus_life_threshold(score)` in `game.py`. Award one bonus life whenever the score reaches a multiple of 1500: 1500, 3000, 4500, etc. Make sure the same threshold does not award multiple lives accidentally. Do not modify the other three features.

Result / implementation

The threshold function was implemented to return 1500, using the existing bonus-award logic for successive thresholds.

Relevant code

```
def bonus_life_threshold():  
    return 1500
```

Scope: This document intentionally excludes Python/Pygame setup, Git troubleshooting, video recording, and submission guidance.